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- So **KKT** condition is **not** a necessary condition for a **local minimum**, although FJ conditions are **necessary conditions** for a **local minimum.**

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- $\mathbf{c}^T - \mathbf{y}^T A - \mathbf{v}^T = \mathbf{0}$, or $\mathbf{c}^T - \mathbf{y}^T A = \mathbf{v}^T$
  and $v_i x_i = 0$ for all $i = 1, ..., n$.
  where $\mathbf{v} = [v_1, \ldots, v_n]^T$ is a non negative vector,
  and $\mathbf{y} = [y_1, \ldots, y_m]^T$ is unrestricted in sign.
- The above conditions are equivalent to the **dual feasibility** and the **complementary slackness** conditions for the above LPP.

- **Exercise:** The KKT conditions for the linear programming problem (LPP):
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- **Exercise:** What are the KKT conditions for the following linear programming problem
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  subject to, $A\mathbf{x} \geq \mathbf{b}, \mathbf{x} \geq \mathbf{0}$.

- Consider the problem (P) of minimizing $f(x_1, x_2) = -x_1 x_2 + x_1^2 + 2x_2^2 - 2x_1 + e^{x_1+x_2}$ over R^2.

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- Minimize $(x_1 - 2)^2 + (x_2 - 3)^2$
  subject to
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  3. Does there exist an $\mathbf{x}^* \in Fea(P)$ such that $G_{0,\mathbf{x}^*} = D_{\mathbf{x}^*}$?
  4. Does there exist an $\mathbf{x}^* \in Fea(P)$ such that $G_{0,\mathbf{x}^*} \neq D_{\mathbf{x}^*}$?
  5. Does there exist an FJ point which is not a KKT point?

- Consider the following problem:
  Minimize $4x_1^2 - x_2^2 + 8x_1 x_2$
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  1. Does $[0,0]^T$ satisfy the first and the second order necessary conditions for a local minimum?
  2. Find all points $\mathbf{x}^*$ in the feasible region at which $G_{0,\mathbf{x}^*} = \phi$.
- If $\mathbf{x}^* \in S$ is such that for all $\mathbf{d} \in D_{\mathbf{x}^*} (\neq \phi)$, $\nabla f(\mathbf{x}^*)\mathbf{d} > 0$ then does it imply that $\mathbf{x}^*$ is a local minimizer of f?

- Consider the following problem:
  Minimize $-x_1^2 - 4x_1x_2 - x_2^2$
  subject to
  $x_2^2 + x_1^2 = 1.$
  1. If possible find an FJ point which is not a KKT point.

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  2. If possible find a KKT point which is not an optimal point.
  3. Are the first and the second order necessary conditions for a local minimum satisfied at the KKT point/s?
  4. If the objective function is changed to $2x_1^2 - x_1x_2 + x_2^2 - x_2$ and if $S = Fea(P) = \{(x_1, x_2) : x_2^2 + x_1^2 \leq 1\}$ then find all optimum solutions to this problem.

- For a nonlinear programming problem (P) of the form,

$$\begin{aligned} &\text{Minimize } f(\mathbf{x}) \\ &\text{subject to } g_i(\mathbf{x}) \leq 0, \text{ for } i = 1, \ldots, m, \quad \mathbf{x} \in \mathbb{R}^n, \end{aligned}$$

where all the g_i's and f are continuously differentiable throughout $\mathbb{R}^n$, check the correctness of the following statements with proper justification.

1. If $\mathbf{x}^* \in Fea(P)$ is a KKT point of the above problem then $-\nabla f(\mathbf{x}^*)$ lies in the cone generated by $\nabla g_i(\mathbf{x}^*), i \in I$, where I gives the indices of the binding constraints (given by g_i's) at $\mathbf{x}^*$.

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3. If $\nabla f(\mathbf{x}^*) = 0$ for some $\mathbf{x}^* \in Fea(P)$ then $\mathbf{x}^*$ is a KKT point.
4. If $\mathbf{x}^*$ is an FJ point and there is a solution to the FJ conditions at $\mathbf{x}^*$ with $u_0 = 0$ (u_0 is the coefficient of $\nabla f(\mathbf{x}^*)$ in the FJ conditions), then $\mathbf{x}^*$ is not a KKT point.

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- If all the g_i's and f are convex functions and $\mathbf{x}^*$ is such that $F_{0,\mathbf{x}^*} \cap G_{0,\mathbf{x}^*} = \phi$, then $\mathbf{x}^*$ is a global minimum of f in S.

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- If $\mathbf{x}^*$ is a local minima of (P) with $F_{0,\mathbf{x}^*} \neq \phi$ but $G_{0,\mathbf{x}^*} = \phi$ then $\mathbf{x}^*$ is not a KKT point.

- There exists a (P) such that for all $\mathbf{x}^* \in S$, $G_{0,\mathbf{x}^*} = D_{\mathbf{x}^*} \neq \phi$.

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- There exists a (P) with $Fea(P) \neq \phi$ such that $D_{\mathbf{x}^*} = \phi$ for all $\mathbf{x}^* \in S$.
- Give examples of nonconstant functions on $\mathbb{R}^n$ which are both convex and concave and those which are neither convex nor concave.

- For a linear programming problem (P) of the form,
  Minimize $\mathbf{c}^T\mathbf{x}$ subject to $A_{m\times n}\mathbf{x} \leq \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$,
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  Solve the KKT conditions for u_i's. Hence find an optimal solution of the dual.

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- Minimize $-x_1$
  subject to
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  $-(1 - x_1)^3 - x_2 = 0.$
- Check whether $[1, 0]^T$ is a KKT point of the above problem. How many feasible points does this problem have? What is your conclusion?